Threat Hunting: Find the Story in the Logs

A beginner notebook for forming a hypothesis, exploring synthetic events, and writing evidence-based findings

NOTEBOOK STYLE Handwriting-inspired typeset notes for printing and revision; not a scan of actual handwriting.

1. What is threat hunting?

Threat hunting is a structured search for suspicious patterns that routine alerts may not have surfaced. A hunter starts with a question, examines available telemetry, compares activity with a normal baseline, and documents what the evidence does and does not show.

THE MINDSET A pattern is a clue, not proof. Keep facts, assumptions, and unknowns separate.

2. A simple hunting loop

1. Ask a focused question, such as "Do several failed logins precede a successful login in this sample?"
2. Choose relevant records and fields: time, user, host, source address, event type, result.
3. Set a clear time window and sort events chronologically.
4. Compare the pattern with the expected baseline and nearby context.
5. Validate the pattern against other benign explanations.
6. Write a short finding with evidence, confidence, limitations, and a next check.

3. Read the sample fields

Field	Plain-language meaning
timestamp_utc	When the training event was recorded (UTC).
host	Synthetic device name, not a real asset.
user	Synthetic account label.
event_type	The kind of activity represented.
source_ip	Reserved documentation IP space used in training data.
result	Success, failure, allowed, or blocked.
label	Teaching label to support discussion; not a real incident verdict.

4. Practice: investigate a login pattern

Open threat-hunt-auth-dns-samples.csv. Every row is fabricated. Look for one account with several failures followed by a success, then check time, host, and source context.

1. Record the case IDs and timestamps that support your hypothesis.
2. Check whether the account and host appear elsewhere in the same sample.
3. Write one plausible benign explanation and one alternative explanation.

4. Identify at least two additional data sources you would seek in a real, authorised investigation.
5. State whether the evidence supports, weakens, or leaves the hypothesis unresolved.

5. Query-thinking example

The following KQL-shaped example illustrates the logic for a Microsoft Sentinel-style table. Column and table names vary by data source, so adapt it only inside an authorised lab with matching fields.

```
SecurityEvent
| where EventID in (4624, 4625)
| summarize failures=countif(EventID == 4625), successes=countif(EventID == 4624) by Account, IpAddress
| where failures >= 5 and successes > 0
```

IMPORTANT This example finds a pattern worth reviewing; it cannot tell you by itself whether an account was compromised.

6. Finding template

Write down	Prompt
Question	What did I try to find?
Evidence	Which event IDs, times, hosts, users, and fields support the finding?
Alternative	What legitimate activity could explain the pattern?
Confidence	Low / medium / high, with a reason.
Unknowns	What data is missing or incomplete?
Next check	What is the safest, most useful authorised follow-up?

Further learning

- MITRE ATT&CK knowledge base
- Microsoft threat hunting overview
- KQL overview